

Forcing-agent attribution: how the residual and its decomposition are computed

CESM2 diffusion emulator, checkpoint ep0860

Definitions

Let $s \in \{c, m\}$ index the side — c the CESM2 reference, m the emulator — and let the experiment appear directly as a subscript, $X \in \{\text{ssp370}, \text{ghg}, \text{aaer}\}$:

- **ssp370 — all-forcing.** Every agent evolves together (greenhouse gases, aerosols, ozone, land use, biomass burning, solar, volcanic). Historical to 2014, SSP3-7.0 after; the window below sits entirely on the SSP3-7.0 side.
- **ghg — GHG-only.** A single-forcing run in which only greenhouse gases vary; everything else is held at its 1850 value.
- **aaer — aerosol-only.** Only anthropogenic aerosols vary, everything else at 1850.

The two single-forcing ensembles follow SSP3-7.0 after 2015 as well, so all three experiments share one scenario pathway — which is what makes their difference meaningful — and all three end at 2050, which is where the single-forcing archive stops.

The averaging window is $\mathcal{W} = [2041, 2050]$: after 2015, so no major volcanic eruption falls inside it, and before 2050.

The evaluation files store per-member anomaly maps $x_{X,i}^s(t, \phi, \lambda)$, each referenced to its own side's 1850–1900 climatology, for member $i = 1, \dots, N_X^s$.

Symbols

Symbol	Value here	Meaning
ϕ, λ	192×288	latitude, longitude on the native CESM2 grid
t	years	time index, one value per year
$\mathcal{W}$	[2041, 2050]	the averaging window; $ \mathcal{W} = 10$ years
s	c or m	the side: c = CESM2 reference, m = emulator
X	ssp370, ghg, aaer	the experiment, named as above
i	$1 \dots N_X^s$	ensemble member index
N_X^s	3, 10, 10	members for ssp370, ghg, aaer; equal on both sides after --match-members
$x_{X,i}^s$	K	a member's anomaly map, referenced to that side's own 1850–1900 climatology
$\bar{x}_{X,i}^s$	K	Eq. (1): one member averaged over $\mathcal{W}$
$\langle x_X^s \rangle$	K	Eq. (2): the ensemble mean of those, a single map
R^s	K	Eq. (3): the residual — what the two agents fail to explain between them
$w(\phi)$	$\cos \phi$	area weight; cells shrink towards the pole
$\langle\langle \cdot \rangle\rangle$	—	Eq. (4): area-weighted mean over $\phi \geq 60^\circ\text{N}$, i.e. the Arctic
b_1	+1.2180 K	first bar: CESM2's Arctic-mean residual
b_5	+2.1416 K	last bar: the emulator's
Δ_{ssp370}	+0.4328 K	middle bar: emulator minus CESM2 for ssp370, entering +
Δ_{ghg}	+0.2354 K	middle bar: emulator minus CESM2 for ghg, entering –
Δ_{aaer}	+0.2554 K	middle bar: emulator minus CESM2 for aaer, entering –

A bar over a symbol always means a mean over time, angle brackets a mean over members, and double angle brackets a mean over Arctic area. The Δ terms are emulator errors; R is not an error but a property of the climate response, present in CESM2 itself.

The calculation

Step 1 — decadal mean per member.

$$\bar{x}_{X,i}^s(\phi, \lambda) = \frac{1}{|\mathcal{W}|} \sum_{t \in \mathcal{W}} x_{X,i}^s(t, \phi, \lambda) \quad (1)$$

Step 2 — ensemble mean.

$$\langle x_X^s \rangle = \frac{1}{N_X^s} \sum_{i=1}^{N_X^s} \bar{x}_{X,i}^s \quad (2)$$

Under equal sampling (**--match-members**) the emulator ensemble is truncated to the reference count, $N_X^m = N_X^c$, so that neither side's mean is better converged than the other's. Here $N_{\text{ssp370}} = 3$, $N_{\text{ghg}} = 10$, $N_{\text{aaer}} = 10$.

Step 3 — the residual field.

$$R^s = \langle x_{\text{ssp370}}^s \rangle - \langle x_{\text{ghg}}^s \rangle - \langle x_{\text{aaer}}^s \rangle \quad (3)$$

R^s is what the two agents fail to explain between them. It is not a pure interaction term: the single-forcing runs hold ozone, land use, biomass burning, solar and volcanic forcing at 1850, so $R = (\text{other forcings}) + (\text{nonlinear interaction})$.

Step 4 — the Arctic operator. With area weights $w(\phi) = \cos \phi$, renormalised inside the mask $\phi \geq 60^\circ\text{N}$,

$$\langle\langle f \rangle\rangle = \frac{\sum_{\phi \geq 60^\circ} \sum_{\lambda} w(\phi) f(\phi, \lambda)}{\sum_{\phi \geq 60^\circ} \sum_{\lambda} w(\phi)} \quad (4)$$

The five bars

The two end bars are the Arctic-mean residual on each side,

$$b_1 = \langle\langle R^c \rangle\rangle = +1.2180 \text{ K}, \quad b_5 = \langle\langle R^m \rangle\rangle = +2.1416 \text{ K}. \quad (5)$$

The three middle bars are the emulator's bias for one experiment, carrying the sign with which that experiment enters Eq. (3):

$$\Delta_{\text{ssp370}} = +\langle\langle \langle x_{\text{ssp370}}^m \rangle - \langle x_{\text{ssp370}}^c \rangle \rangle\rangle = +(4.674 - 4.242) = +0.4328 \text{ K}, \quad (6)$$

$$\Delta_{\text{ghg}} = -\langle\langle \langle x_{\text{ghg}}^m \rangle - \langle x_{\text{ghg}}^c \rangle \rangle\rangle = -(4.021 - 4.257) = +0.2354 \text{ K}, \quad (7)$$

$$\Delta_{\text{aaer}} = -\langle\langle \langle x_{\text{aaer}}^m \rangle - \langle x_{\text{aaer}}^c \rangle \rangle\rangle = -(-1.488 + 1.233) = +0.2554 \text{ K}. \quad (8)$$

The sign flip is the substance of the panel: the emulator's ghg run is too *cool* and its aaer run too *cold*, yet both terms are *subtracted* in Eq. (3), so both push the residual *up*. Three biases of different physical sign act on R in the same direction.

Arctic mean (K)	CESM2	emulator	bias
ssp370 (all-forcing)	+4.242	+4.674	+0.433
ghg (GHG-only)	+4.257	+4.021	-0.235
aaer (aerosol-only)	-1.233	-1.488	-0.255
Residual R	+1.218	+2.142	+0.924

Why the waterfall closes exactly

$$\langle\langle R^m \rangle\rangle - \langle\langle R^c \rangle\rangle = \Delta_{\text{ssp370}} + \Delta_{\text{ghg}} + \Delta_{\text{aaer}} \quad (9)$$

$$1.2180 + 0.4328 + 0.2354 + 0.2554 = 2.1416$$

Equation (9) is an identity, not a fit. Every operator in the chain — the decadal mean, the ensemble mean, the combination in Eq. (3), and the area-weighted mean — is linear, so they commute and

$$\langle\langle R^m \rangle\rangle - \langle\langle R^c \rangle\rangle = \langle\langle (\langle x_{\text{ssp370}}^m \rangle - \langle x_{\text{ssp370}}^c \rangle) - (\langle x_{\text{ghg}}^m \rangle - \langle x_{\text{ghg}}^c \rangle) - (\langle x_{\text{aaer}}^m \rangle - \langle x_{\text{aaer}}^c \rangle) \rangle\rangle \quad (10)$$

Verified numerically, the closure error is 1.2×10^{-10} K, i.e. floating point. The panel therefore does not test anything: it *attributes* an already-known 0.924 K gap to its three sources, and the finding is that the split is roughly even rather than one bad scenario.

Caveat. After matching, Δ_{ssp370} rests on three members per side, all the CESM2 ssp370 reference in these files provides — and that reference is the 3-member CMIP6 ensemble, whereas ghg and aaer are 10-member LENS2 single-forcing runs. Part of R^c could therefore be a model-configuration difference rather than a forcing interaction. It is both the largest bar and the least certain, and it is the one that would move were the reference rewired to the ten held-out LENS2 ssp370 members. Bar labels in the figure are rounded to two decimals, so reading them off gives 2.15 rather than 2.14; the underlying values close exactly.